WIRELESS NETWORK · SITE ASSESSMENT

Wi-Fi Signal Strength Assessment

Residential coverage review — preliminary assessment

Prepared 24 October 2026 · Version 1.0 · Prepared with WiFiman

PRELIMINARY

REPORT DETAILS

PREPARED	24 October 2026	**MEASUREMENT APP**	WiFiman
COVERAGE AREA	Seven locations: 4 rooms, washroom, hallway, kitchen	**METRIC**	RSSI, in dBm
DOCUMENT STATUS	Preliminary — illustrative estimates only		

1. Executive summary

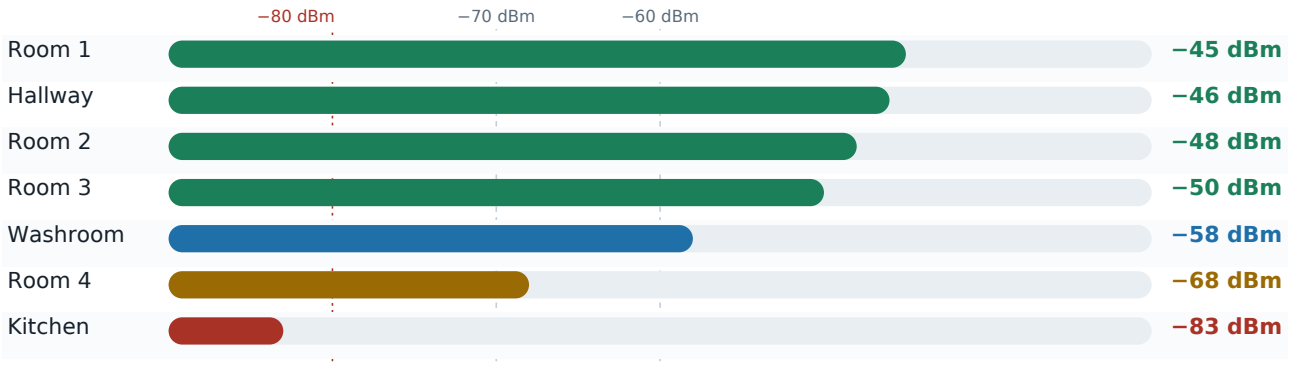
This report sets out the expected Wi-Fi signal strength at seven locations within a single dwelling. Reception is expected to be **excellent** in Rooms 1–3 and the hallway (−45 to −50 dBm), **good** in the washroom (−58 dBm) and **fair** in Room 4 (−68 dBm). The **kitchen is the exception**, at an estimated −83 dBm, which is below the threshold for dependable connectivity.

The kitchen is therefore the highest-priority location for remedial work, and Room 4 should be re-checked once any router relocation has been completed. Because the figures are estimates rather than measured readings, Section 6 sets out how to convert them into a verified survey.

7	4	1	38 dB
locations assessed	rated excellent	rated poor — kitchen	spread across areas

2. Coverage profile

Estimated signal strength by location, ordered from strongest to weakest. The kitchen sits below the −80 dBm threshold at which connections typically become unreliable.



Scale: −90 dBm (weak, left) to −30 dBm (strong, right). Dashed guides mark the −60, −70 and −80 dBm rating thresholds.

3. Area-by-area assessment

AREA	RSSI	RATING	RELATIVE STRENGTH	INTERPRETATION
Room 1	−45 dBm	Excellent		Strong signal expected throughout the room.
Room 2	−48 dBm	Excellent		Strong signal expected throughout the room.
Room 3	−50 dBm	Excellent		Strong signal expected throughout the room.
Room 4	−68 dBm	Fair		Usable, but likely to be slower or less stable.
Washroom	−58 dBm	Good		Reliable for everyday use; some speed loss possible.
Hallway	−46 dBm	Excellent		Strong signal expected along the full length.
Kitchen	−83 dBm	Poor		Unreliable connectivity expected; dropouts likely.

RSSI is expressed in dBm; values closer to zero indicate a stronger signal, so −45 dBm is stronger than −83 dBm. The bars above are scaled across the −30 to −90 dBm range and are intended for comparison only.

4. Rating guide

The following practical ranges are used throughout this report. They describe the expected end-user experience at the stated signal level.

RATING	RSSI RANGE	TYPICAL EXPERIENCE
Excellent	−30 to −50 dBm	Full-speed connectivity for streaming, video calls and file transfer.
Good	−51 to −60 dBm	Reliable everyday performance; throughput begins to tail off at range.
Fair	−61 to −70 dBm	Browsing and messaging fine; video calls and large transfers may stutter.
Weak	−71 to −79 dBm	Unstable connection; dropouts likely, particularly on the 5 GHz band.
Poor	≤ −80 dBm	Frequent disconnections; some devices may fail to stay connected.

Actual throughput is also influenced by interference, the Wi-Fi band in use (2.4 GHz reaches further; 5 GHz is faster), the capability of the router and client device, building materials and the number of connected devices.

PRIORITY GROUPING

ACT NOW

Kitchen

Below the reliable-service threshold; remedial action required.

MONITOR

Room 4

Serviceable, but close to the weak boundary. Re-test after changes.

NO ACTION

Rooms 1–3, washroom, hallway

Expected to meet normal household requirements without changes.

5. Recommended actions

The actions below are listed in the order in which they should be carried out. Steps 1, 2 and 4 can be completed at no cost; step 3 may require additional hardware.

01 Relocate the router

HIGH PRIORITY

Place the router in a central, elevated and open position, clear of masonry, metal surfaces and large appliances such as the refrigerator and microwave. Router placement is the most common cause of a weak signal at the far end of a property.

02 Re-test the kitchen and Room 4

HIGH PRIORITY

Measure signal where devices are actually used, not at the doorway. Record the reading at seated or standing height in each location and repeat the test at two different times of day to account for network load.

03 Extend coverage if required

CONDITIONAL

If a verified kitchen reading remains at or below −80 dBm after relocation, add a mesh node or access point positioned roughly midway between the router and the kitchen. A wired (Ethernet-backhaul) access point is preferable where cabling is practical.

04 Capture verified readings in WiFiman

HIGH PRIORITY

Log RSSI for every location together with the device, the Wi-Fi band (2.4 GHz or 5 GHz) and the date and time. Repeat the survey after any change to router placement or equipment so that before-and-after results can be compared directly.

6. Measurement note and limitations

Basis of the figures

No on-site RSSI readings were supplied for this assessment. WiFiman is the application identified for the survey, and the dBm values in Section 3 are representative estimates chosen to reflect the reported signal category for each location. They are **not** readings exported from WiFiman and should not be relied upon as measured data or used as the basis for a contractual commitment.

To convert this into a verified survey:

- In WiFiman, record RSSI at each location and note the Wi-Fi band, the test device and the date and time of each reading.
- Take readings at the height and position where devices are normally used, and repeat each measurement at least twice to identify variation.
- Note any household appliances, mirrors, metal surfaces or thick walls between the router and each location; these commonly explain a localised drop in signal.
- Repeat the survey after any change to router placement or equipment and compare the results against the baseline in Section 3.

Appendix A — survey record template

Use this sheet to record verified WiFiman readings during the follow-up survey. Complete one row per location, then compare the results against the estimates in Section 3. Retain the completed sheet with this report so that any change to router placement or equipment can be evaluated against a documented baseline.

LOCATION	RSSI (dBm)	BAND	TEST DEVICE	DATE / TIME
Room 1				
Room 2				
Room 3				
Room 4				
Washroom				
Hallway				
Kitchen				

ADDITIONAL OBSERVATIONS

Record the band in use (2.4 GHz or 5 GHz) separately where both are broadcast — a location may read well on 2.4 GHz and poorly on 5 GHz. Note the test device as well, since antenna design and Wi-Fi generation vary considerably between phones and laptops.

SURVEY SIGN-OFF

Surveyed by		Signature	
Date of survey		Readings attached	